

Financial Fraud Detection Report

1. Introduction

The Financial Fraud Detection System is an end-to-end machine learning solution designed to monitor financial transaction flows, detect anomalous behavior, and flag high-risk transactions. Operating on an evaluation dataset of 5,000 transactions totaling ₹395,903.32, the system successfully flags high-risk transactions to mitigate direct financial loss while balancing investigative workload.

2. Core Portfolio & Operational Metrics

Metric	System Value	Contextual Interpretation
Total Transactions Analysed	5,000	Production test slice (spanning Jan 2023 – Feb 2024).
Total Transaction Value	₹ 3,95,903.32	Overall gross transaction volume processed.
Actual Fraudulent Transactions	482 (9.64%)	Ground-truth confirmed fraudulent activities.
Actual Fraud Exposure Value	₹ 39,520.96	Direct monetary exposure from fraudulent activities.
Model Flagged Transactions	1,284 (25.68%)	Total volume flagged for investigation.
Model Flagged Value	₹ 1,12,920.07	28.52% of total transaction value flagged.
High / Medium / Low Risk Volume	1,284 / 438 / 3,278	25.7% High Risk, 8.8% Medium Risk, 65.6% Low Risk.

3. Model Architecture & Evaluation Performance

While benchmark models (Logistic Regression, KNN, SVC, Decision Trees) provided baseline comparisons during exploratory phases, the final deployed production model is **XG Boost** operating at an optimized classification threshold of **0.69**.

- **Classification Threshold Justification:** Rather than using the default 0.50 cutoff, tuning the threshold to **0.69** prioritizes **Recall (85.23%)** over **Precision (29.07%)**, ensuring 85%+ of actual fraud cases are captured before loss realization.
- **Test Set Confusion Matrix Results:**
 - **True Negatives (TN):** 729
 - **False Positives (FP):** 183
 - **False Negatives (FN):** 13
 - **True Positives (TP):** 75
- **Performance Summary:**
 - **Accuracy:** 80.40%
 - **Precision:** 29.07%
 - **Recall:** 85.23%
 - **F1 Score:** 43.35%
 - **ROC-AUC:** 0.9021
 - **PR-AUC:** 0.4349

4. Key Risk Drivers & Behavioral Explainability

- **Geographic & Transaction Type Vulnerability:** International transactions represent only **8.82%** of total transaction volume (441 transactions) but exhibit an alarming **96.37% predicted fraud rate** (425 flagged).
- **Temporal Patterns:** Peak fraud activity concentrates heavily during late-night and early-morning hours (12:00 AM – 5:00 AM), where flagged rates consistently hover between **55% and 70%**, sharply declining to ~10% during daytime hours.

□ **Payment Method Breakdown:**

- PayPal: 28.18% Flagged Rate
- UPI: 27.78% Flagged Rate
- Debit Card: 27.48% Flagged Rate
- Net Banking: 25.65% Flagged Rate
- Credit Card: 19.14% Flagged Rate

□ **Primary Explainability Indicators:**

- *Night-time transaction*: 1,043 instances
- *International transaction*: 312 instances
- *Suspicious keyword detected*: 260 instances
- *High transaction amount deviation*: 189 instances
- *International + night-time compound flag*: 107 instances

5. Operational Workflow & Investigation Decision Guidelines

- **Immediate Automated Holds (High Priority / High Risk):** All 1,284 transactions scoring above the 0.69 threshold (average fraud probability: **73.50%**) route directly into the priority queue. Mandatory two-factor re-authentication or transaction freezes should trigger automatically on international night-time transactions.
- **Secondary Review Queue (Medium Risk):** The 438 transactions sitting in the medium band (risk scores 30–60) should undergo passive behavioural biometric checks and velocity checks rather than hard stops to prevent user drop-off.
- **Continuous Monitoring:** Regularly retrain XG Boost against emerging transaction types and implement real-time streaming pipelines to lower false positive rates without sacrificing the 85.2% recall rate.

6. Data Preprocessing & Feature Engineering

- **Handling Missing Values:** No missing values were found in the dataset.
- **Scaling Features:** Robust Scaler was utilized to scale numerical features, specifically to mitigate the effect of skewness and extreme outliers (such as on scaled amount and scaled time).
- **Balancing the Dataset:** To address the severe class imbalance (where fraud cases accounted for only 0.172% of the data), synthetic sampling and resampling strategies—specifically SMOTE (Synthetic Minority Over-sampling Technique) and under sampling—were applied.
- **Dimensionality Reduction & Visualization:** Both PCA (Principal Component Analysis) and t-SNE were employed for dimensionality reduction to visualize transaction clusters and class separability in reduced dimensional space.
- **Feature Representations:** The dataset utilizes transformed PCA components (V_1 through V_{28}) alongside engineered/scaled features (scaled_amount, scaled_time), which feed directly into the downstream classification pipelines.

7. Machine Learning Models Used

- **XG Boost (Extreme Gradient Boosting):** Selected as the final model for the system and dashboard. It operates at an optimized decision threshold of 0.69 (rather than 0.50) to optimize fraud recall (85.23%) and fraud probability scoring.
- **Logistic Regression:** Used as a linear, interpretable baseline for classification and probability benchmarking.
- **K-Nearest Neighbors (KNN):** Evaluated for pattern-based fraud detection across transaction feature spaces.
- **Support Vector Classifier (SVC):** Tested for handling non-linear relationships in high-dimensional feature representations.

- **Decision Tree:** Implemented as a rule-based model to evaluate simple decision boundaries and explainability.
- **Random Forest:** Implemented as an ensemble tree baseline to reduce variance and improve accuracy.

8. Model Evaluation & Performance Metrics

- **Accuracy:** Measures overall correctness of predictions across all transactions.
- **Precision:** Evaluates the proportion of flagged transactions that were fraudulent (evaluating false positives).
- **Recall:** Measures fraud detection sensitivity—the proportion of actual fraudulent transactions captured by the model.
- **ROC-AUC Score:** Assesses fraud classification performance by measuring the area under the Receiver Operating Characteristic curve across varying probability thresholds.

9. Confusion Matrix Analysis

The initial experimental phase evaluated models on a resampled test split to compare baseline error distributions:

- **Logistic Regression:**
 - **True Negatives (TN):** 91
 - **False Positives (FP):** 3
 - **False Negatives (FN):** 11
 - **True Positives (TP):** 85
 - **Outcome:** High specificity with a small false positive footprint (3 cases) and an 88.5% recall on the resampled subset.

- **K-Nearest Neighbors (KNN):**
 - **True Negatives (TN):** 90
 - **False Positives (FP):** 4
 - **False Negatives (FN):** 14
 - **True Positives (TP):** 82
 - **Outcome:** Slightly higher false negative rate (14 missed fraud cases) compared to Logistic Regression.

Production Model Confusion Matrix

On the production test dataset of 1,000 transactions, the final **XG Boost** model was evaluated at an operational decision threshold of **0.69**.

10. Conclusion & Future Work

Current Project Conclusions:

Successful Implementation: Machine learning models were implemented to classify and detect fraudulent financial transactions from highly imbalanced transaction data.

Benchmark Performance: During the exploratory modelling phase across resampled baselines, the best-performing baseline models achieved an ROC-AUC score of 0.98 (KNN) to 0.99 (Logistic Regression, SVC).

Production Deployment: The finalized production system deployed an XG Boost model calibrated at a tuned classification threshold of 0.69, successfully achieving 85.23% Recall and an ROC-AUC of 0.9021 to catch high-impact fraud while managing queue volumes.

Risk Factor Identification: High-risk indicators were successfully isolated, uncovering that international transactions (96.37% flagged rate) and late-night transaction windows (12:00 AM – 5:00 AM) represent the highest operational fraud vulnerabilities.

Future Work & Roadmap:

Deep Learning Architecture: Explore sequential and recurrent deep learning architectures, such as Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNNs), to capture temporal sequences and evolving transaction histories over time.

Real-Time Streaming Pipelines: Transition from batch scoring to real-time fraud detection engines (e.g., streaming inference via Kafka/Flink) to block fraudulent transactions at the authorization gateway prior to settlement.

Hyperparameter Optimization: Conduct automated hyperparameter tuning (e.g., Bayesian optimization, Optuna) on ensemble models to lower the false positive rate while preserving high recall.

Dynamic Thresholding: Implement adaptive, context-aware thresholding that dynamically shifts the decision cutoff based on geography, payment channel, or transaction value rather than applying a static 0.69 cutoff.

Graph & Network Analysis: Incorporate Graph Neural Networks (GNNs) to detect mule networks, shared device fingerprints, and synchronized fraud rings across multiple accounts.